

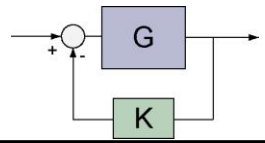
ME 2801

Introduction to Control Systems

W1: Course Introduction



Plan of the Day



Introductions

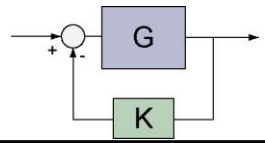
Course Overview

Next Steps

- Reading for next time



Instructor Intro.



Professor Brian Bingham

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- Phone: (831) 656-2396

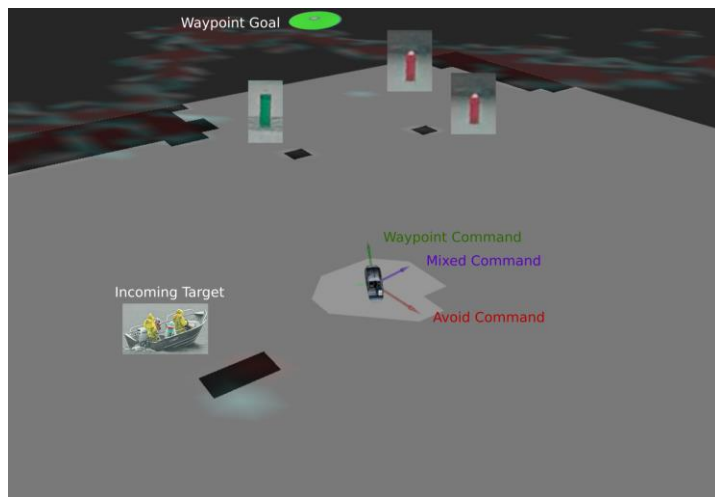
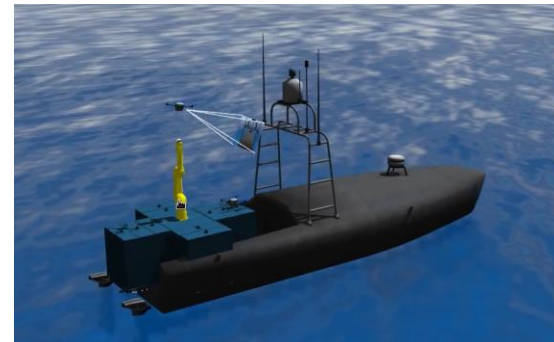
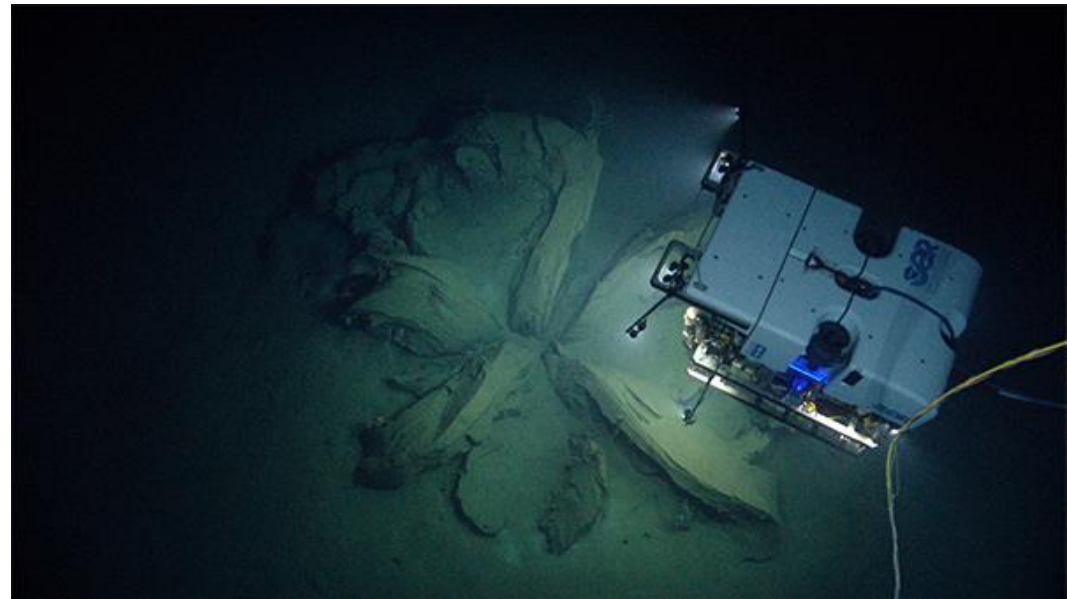
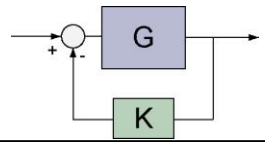
Biographical Information

- PhD in ME, 2003, MIT
- 1 Year at Woods Hole Oceanographic
- 4 Years at Franklin W. Olin College of Engineering
- 6 Years at University of Hawaii
- 10 Years at NPS

Research Interests

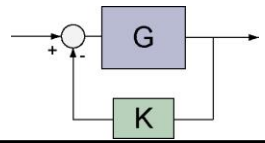
- Dynamics, controls, estimation, navigation
- Marine robotics

Research Interests





Class Introductions



Name?

Where are you from?

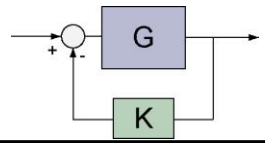
What was your last job?

What are you studying at NPS?

What is your next job?



Course Overview

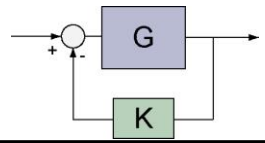


Introduction to Control Systems

- Develop a set of analytical/mathematical tools
 - Laplace transforms, transfer functions, time-domain and s-domain metrics, root locus, etc.
- Apply these tools to model simple systems
- Design classical feedback controllers (PID, lead/lag) to improve the dynamic stability and performance

Goals: After completing this course you should...

- Know what feedback is and why it might be used
- Understand the basic vocabulary of feedback control
- Be on friendly terms with MATLAB
- Have designed your own PID compensators
- Know if “Robotics and Controls” might be a specialization track for you.



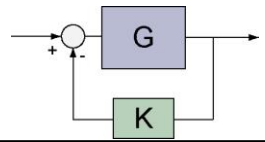
Control of an Uncrewed Surface Vessel (USV)

- **Lab 1: System ID**
 - Build a model to predict USV dynamics
- **Lab 2: PID Control**
 - Tune surge and yaw control
 - Teach the USV to go straight.
- **Lab 2: Navigation**
 - Tune waypoint navigation
 - Teach USV to complete simple missions





Course Plan



What we will do...

Assignments	How Many	Weight
Reading Quizzes	~7	10%
Homework	~7	30%
Exams	1	30%
Labs	3	30%

When... ([Course Wiki](#))

Pages / ME2801: Introduction to Control Systems  

 Edit  View inline comments  Save for later  Watching  Share ...

Schedule (26-3)

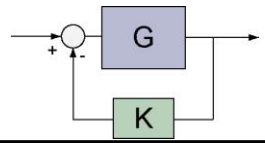
Created by Bingham, Brian (CIV), last modified just a moment ago

**** Unless explicitly scheduled, the only synchronous class meetings will be Mondays and Thursdays**

Week	Monday - In-Person	Tuesday - Asynchronous **	Wednesday - Asynchronous	Thursday
1: 30 Mar	Course Introduction and Laplace Introduction Slides: lec_w1_intro.pdf	No In Person Meeting Required Videos (54 min): - Laplace Transforms Laplace Introduction (11:27) Laplace Properties and Pairs (13:58) - Laplace Transform Examples: Part 1: Simple examples (7:22) Part 2: Higher order systems (6:00)	No In Person Meeting See Tuesday video lectures	Before Class: - Watch videos - Start HW1 Friday: Due by 1700 Friday: - HW1 - See Assignments (25-3) for assignment document.



Grading

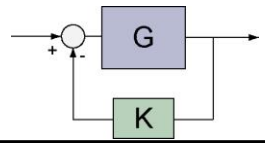


Homework

- Goals:
 - Incentivize keeping up with course material
 - Rapid debugging of problems
- Method:
 - Collected weekly
 - “Solutions” are provided (Hazard!)
 - Students highlight problem areas or confusion
 - Credit/No-Credit

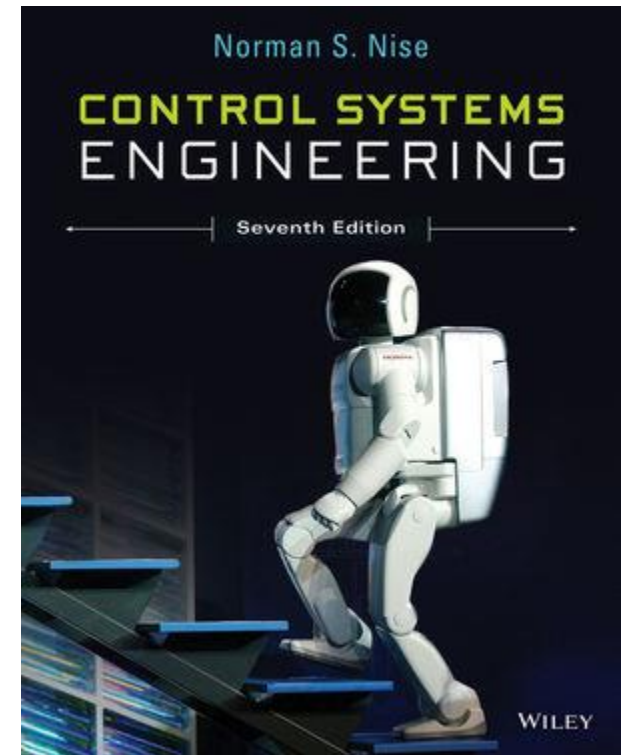
Quizzes

- Goal: Determine if you did the reading
- Graded in class
- Grade Determination:
 - Drop lowest score
 - Avg. $> 70\%$ $\rightarrow$ Full Credit
 - $40\% > \text{Avg.} > 69\%$ $\rightarrow$ $\frac{3}{4}$ Credit
 - Avg. $< 40\%$ $\rightarrow$ No Credit



Textbook

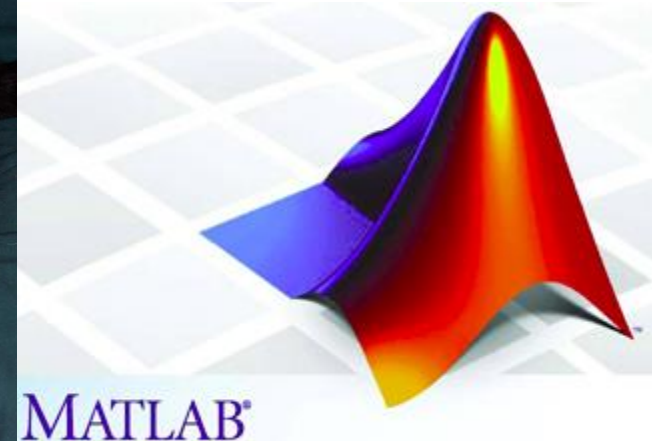
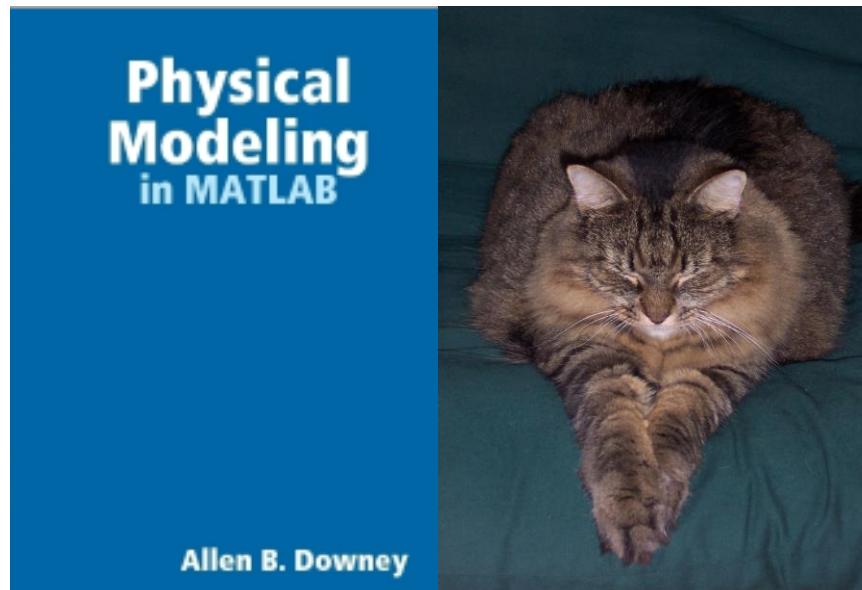
- Reading assignments
- Reading quizzes



Website / Wiki

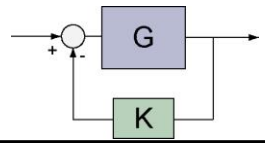
- Calendar, assignments and resources

MATLAB





Policies/Expectations



Collaboration and Generative AI

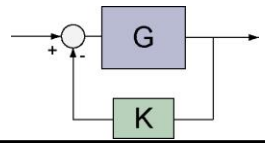
- The goal of this policy is to encourage meaningful learning and prevent activities that hinder your educational development.
- Our guiding principle for academic honesty is: “be reasonable.”
- All submitted work must be your own
- Exams and quizzes will be done individually

Late Work

- Credit for work submitted past the announced deadline will be reduced by 10% per day (or portion of a day) up to 50%. Work will not be accepted more than 5 days late.
- Deadline extensions will not be granted within 24 hours of assignment deadlines without extenuating circumstances.



Policies/Expectations

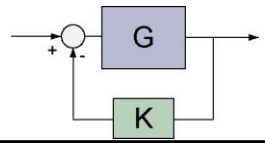


Use of Technology

- We will use e-mail for communication.
I expect everyone to read their email daily
- The website will be updated frequently, roughly once a week, as we move through the course.
I reserve the right to make adjustments to the schedule/calendar/assignments as we go.



Expectations



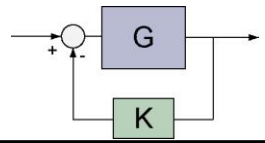
I will...

- Be organized
- Let you know what is coming up
- Provide meaningful feedback
- Be available if you need help (please schedule an appointment)
- Use your time efficiently
- Be excited about the material

I expect the same from you.



Next Steps



Textbook Reading:

- Chapter 2: 2.1-2.3
- Appendix B (MATLAB): ch2p1-ch2p8 (poly and residue functions)
- Take a look at HW1, Due Friday

Tuesday and Wednesday, Asynchronous

- Review content videos

Thursday, In-Person:

- Meeting to review video topics and work on HW1

Friday:

- HW1 Due
- Optional MATLAB refresher